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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

Hospital Patient Registration and Appointment Scheduling System

Submitted by

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Academic Year: **2025 – 2026**

APRIL 2026

Semester: **II**

DECLARATION

I hereby declare that the project entitled **"Hospital Patient Registration and Appointment Scheduling System"** submitted in partial fulfillment of the requirements for the course **Java programming** is a **record of my original work** carried out under the guidance of **Mr. Aravind**, Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

NANDHINI C

CERTIFICATE

This is to certify that the project entitled **"Hospital Patient Registration and Appointment Scheduling System"** is a **bona fide work** carried out by **NANDHINI C**, **[1-'B'/2025-2026], B.Sc.Computer Science KG College of Arts & Science**, in partial fulfillment of the requirements for the course **Javaprogramming**.

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: Coimbatore

Date: April 2026

Submitted for the Viva-Voce Examination held on. /04/2026

Guide:
Mr. Aravind.

Head of the Department
Dr. Boobalan

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ABSTRACT

The Hospital Patient Registration and Appointment Scheduling System is a Java-based application designed to simplify and automate the process of managing patient information and scheduling appointments in hospitals. In many healthcare institutions, patient registration and appointment handling are often done manually, which can lead to long waiting times, data errors, and difficulty in managing records efficiently.

This system provides a digital solution that allows hospital staff to register patients, store their personal and medical details, and schedule appointments with doctors and accurately. It helps in organizing patient data, avoiding duplicate entries, and managing appointment timings effectively. The system also ensures that patients can be assigned to doctors based on availability, reducing waiting time and improving overall service quality.

Developed using Core Java concepts such as classes, objects, methods, and control structures, the application can be implemented as a console-based or GUI-based system. The main objective of this project is to improve efficiency, reduce manual workload, and provide a reliable system for handling patient registration and appointment scheduling. This project demonstrates the practical application of Java programming in developing healthcare management systems.

1. INTRODUCTION

In hospitals, managing patient registration and appointment scheduling is a critical task that directly affects the quality of healthcare services. Traditionally, many hospitals rely on manual methods such as paper records or basic spreadsheets to register patients and schedule appointments. These methods are often time-consuming, prone to errors, and make it difficult to manage large volumes of patient data efficiently.

With the increasing number of patients, there is a need for an automated system that can handle patient registration and appointment scheduling in a fast and organized manner. A computerized system helps in maintaining accurate records, reducing waiting time, and improving the overall efficiency of hospital operations.

1.1 Objective

- To develop a hospital management system using Java
- To register patients digitally
- To schedule appointments efficiently
- To reduce manual errors

1.2 Overview

This system provides a GUI where users can:

- Enter patient details
- Register patients
- Book appointments
- Store records in files

- **1.3 Features**

- User-friendly interface
- Patient registration
- Appointment booking
- File storage
- Error handling

2. PROGRAM ANALYSIS

2.1 Problem Definition

Manual hospital systems lack efficiency and accuracy in managing patient records and appointments.

2.3 Scopes and Limitations

Scope

- Useful for small hospitals
- Easy patient record management

Limitations:

- No database (only file storage)
- No login system

3. SYSTEM DESIGN

System design describes how the Hospital Patient Registration and Appointment Scheduling System is structured and how different components interact with each other. The system is designed using Java Programming Language and GUI components from Java Swing.

The design follows an object-oriented approach using concepts like:

- Inheritance
- Encapsulation
- Polymorphism

The system consists of multiple modules that work together to perform operations efficiently.

3.1 Modules

1**. Patient Registration Module**

Collects patient details such as Name, Age, and Patient ID

Creates a Patient object

Stores the data in a file (patients.txt)

2. **Appointment Booking Module**

Allows users to enter Doctor Name and Appointment Date

Creates an Appointment object

Saves appointment details in a file (appointments.txt)

3. **File Management Module**

Handles storing and retrieving data

Uses FileWriter class for file operations

Implements polymorphism using FileManager and PatientFileManager

4. **GUI Module**

Provides user interface using Java Swing

Contains text fields, buttons, and display area

Handles user interaction through event listeners

5. **Display Module**

Displays registered patient details

Shows appointment booking confirmation

3. METHODOLOGY

The methodology of the Hospital Patient Registration and Appointment Scheduling System describes the step-by-step process followed to design, develop, and implement the project using Java and Java Swing.

4.1. Problem Analysis

In many hospitals, patient registration and appointment booking are handled manually, which leads to delays, errors, and difficulty in maintaining records.

To overcome these issues, a computerized system was analyzed and planned to:

- Store patient details digitally
- Reduce paperwork
- Improve efficiency and accuracy

4.2. System Design

The system was designed using Object-Oriented Programming (OOP) concepts such as:

- Inheritance (Person → Patient)
- Encapsulation (Private data members)
- Polymorphism (Method overriding in FileManager)

4.3. Program Development

The coding was implemented in Java with the following components:

- Classes for Patient and Appointment
- File handling to store records in text files
- Event handling for button actions
- GUI components like JTextField, JButton, JTextArea

The system allows users to:

- Register patients
- Book appointments
- Save data into files

4.4. Result and Documentation

The system successfully performs:

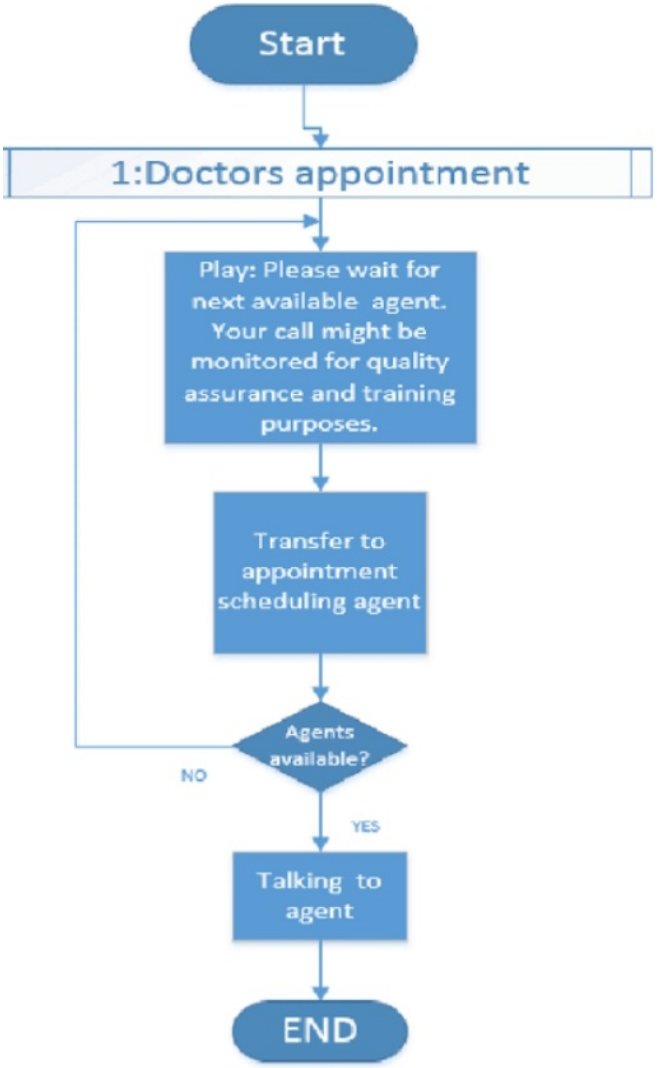
- Patient registration
- Appointment scheduling
- Data storage
- All outputs are displayed on the screen and stored in files.

Documentation was prepared explaining each step clearly.

4.5. Algorithm

1. Start
2. Enter patient details (Name, Age, ID)
3. Click "Register Patient"
4. Save patient details in file
5. Enter doctor name and date
6. Click "Book Appointment"
7. Save appointment details
8. Display output
9. End

4.6. Flowchart



4. SOURCE CODE

```
package hospital;

import javax.swing.*;
import java.awt.*;
import java.awt.event.*;
import java.io.*;

// Base Class (Inheritance)
class Person {
    String name;
    int age;

    Person(String name, int age) {
        this.name = name;
        this.age = age;
    }
}

// Patient Class (Encapsulation + Inheritance)
class Patient extends Person {
    String id;

    Patient(String name, int age, String id) {
        super(name, age);
```

```
        this.id = id;
    }

    String getInfo() {
        return name + "," + age + "," + id;
    }
}

// Appointment Class
class Appointment {
    String doctor;
    String date;

    Appointment(String doctor, String date) {
        this.doctor = doctor;
        this.date = date;
    }

    String getInfo() {
        return doctor + "," + date;
    }
}

// Polymorphism (File Saving)
class FileManager {
    void save(String data) throws IOException {
        System.out.println("Saving...");
    }
}
```

```
class PatientFileManager extends FileManager {
    @Override
    void save(String data) throws IOException {
        FileWriter fw = new FileWriter("patients.txt", true);
        fw.write(data + "\n");
        fw.close();
    }
}

// GUI Class
public class HospitalSystem extends JFrame {

    JTextField nameF, ageF, idF, docF, dateF;
    JTextArea area;

    public HospitalSystem() {

        setTitle("Hospital System");
        setSize(400, 400);
        setLayout(new FlowLayout());
        setDefaultCloseOperation(EXIT_ON_CLOSE);

        add(new JLabel("Name"));
        nameF = new JTextField(10);
        add(nameF);

        add(new JLabel("Age"));
        ageF = new JTextField(5);
```

```
add(ageF);

add(new JLabel("ID"));
idF = new JTextField(8);
add(idF);

add(new JLabel("Doctor"));
docF = new JTextField(10);
add(docF);

add(new JLabel("Date"));
dateF = new JTextField(10);
add(dateF);

JButton regBtn = new JButton("Register");
JButton apptBtn = new JButton("Appointment");

add(regBtn);
add(apptBtn);

area = new JTextArea(10, 30);
add(area);

// Register Patient
regBtn.addActionListener(e -> {
    try {
        Patient p = new Patient(
                nameF.getText(),
                Integer.parseInt(ageF.getText()),
```

```
            idF.getText()
        );

        PatientFileManager fm = new PatientFileManager();
        fm.save(p.getInfo());

        area.append("Patient Added: " + p.getInfo() + "\n");

    } catch (Exception ex) {
        JOptionPane.showMessageDialog(this, "Error in Input!");
    }
});

// Appointment
apptBtn.addActionListener(e -> {
    try {
        Appointment a = new Appointment(
            docF.getText(),
            dateF.getText()
        );

        FileWriter fw = new FileWriter("appointments.txt", true);
        fw.write(a.getInfo() + "\n");
        fw.close();

        area.append("Appointment: " + a.getInfo() + "\n");

    } catch (Exception ex) {
        JOptionPane.showMessageDialog(this, "File Error!");
```

```
            }
        });
    }

    public static void main(String[] args) {
        new HospitalSystem().setVisible(true);
    }
}
```

5. INPUT AND OUTPUT SCREENS

==========

HOSPITAL SYSTEM

==========

Name : Rahul

Age : 23

Patient ID : P105

Doctor : Dr. Kumar

Date : 14-04-2026

REGISTER PATIENT / BOOK APPOINTMENT

STATUS:

✔ Patient Registered Successfully

✔ Data Saved in patients.txt

---

6. DISCUSSION & CONCLUSION

7.1 Discussion

The Hospital Patient Registration and Appointment Scheduling System was developed to improve the efficiency of managing patient data and scheduling appointments in hospitals. The system replaces manual record-keeping with a computerized approach, making it easier to store, update, and retrieve patient information.

During the development of this project, various Core Java concepts such as classes, objects, methods, loops, and conditional statements were applied

7.2 Conclusion

The Hospital Patient Registration and Appointment Scheduling System successfully demonstrates the use of Java programming in developing a practical healthcare management application. The system simplifies the process of patient registration and appointment scheduling by replacing manual methods with an efficient computerized solution.

It helps in maintaining accurate patient records, reducing waiting time, and improving the overall workflow of hospital operations. By automating routine tasks, the system minimizes human errors and enhances the quality of service provided to patients.

7. REFERENCES

Books

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